

Education Platform - Technical Project Handover Documentation

This document provides a comprehensive technical overview of the Education Platform project codebase, detailing its architecture, technology stack, and all fully implemented features across both the frontend and backend systems.

Technology Stack & Dependencies

Frontend (`/client`)

- Framework & Core: React 19, React Router DOM (v7), Vite 6, JavaScript (ES6+ / JSX)
- State Management: Redux Toolkit (@reduxjs/toolkit), react-redux
- Styling & UI: Custom CSS Design System (Glassmorphism, Dark/Light tokens), TailwindCSS (@tailwindcss/vite), React Icons (react-icons)
- HTTP Client: Axios with custom interceptors for authentication and token management
- Media & Content Security: DOMPurify for HTML sanitization, YouTube IFrame Player API for video tracking
- Feedback & Visuals: react-hot-toast notifications, Framer Motion animations, Recharts analytical charts

Backend (`/server`)

- Runtime & Framework: Node.js, Express.js (v4)
 - Database & ORM: MongoDB with Mongoose (v8)
 - Authentication & Security: JSON Web Tokens (JWT), bcryptjs, Express Session, Passport.js (Google OAuth 2.0 & GitHub OAuth 2.0), Helmet, CORS, Express Rate Limit, Sanitize-HTML
 - AI Integration: @google/generative-ai (Google Gemini 3.5 Flash)
 - Document & Asset Generation: PDFKit (Vector PDF rendering), qrcode (Buffer generation), uuid
 - File Storage: Multer for handling file attachments and profile media
 - Email Service: Nodemailer for transactional HTML emails
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1. Authentication Features

- Email & Password Authentication: Full user signup and login with bcryptjs password hashing.
 - Role Assignment: Choice of student or instructor role during user registration.
 - OAuth 2.0 Social Login:
 - Google OAuth 2.0: Integrated via Passport.js (passport-google-oauth20).
 - GitHub OAuth 2.0: Integrated via Passport.js (passport-github2).
 - Dedicated client callback handler (OAuthCallback.jsx) for seamless session token reception.
 - JWT Session Management: Bearer token authentication attached to requests via Axios interceptors.
 - Password Recovery:
 - Tokenized password reset link generation and email delivery via Nodemailer (ForgotPassword.jsx).
 - Secure token validation and password updates (ResetPassword.jsx).
 - Route Security Guards:
 - ProtectedRoute component protecting private application pages.
 - Role-based authorization wrappers restricting access to Admin or Instructor routes.
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2. Student Features

- Course Exploration & Discovery (CourseExplore.jsx):
 - Search courses by title/description keyword.
 - Filter courses by category, difficulty level (Beginner, Intermediate, Advanced), and pricing (Free vs. Paid).
 - Detailed course landing pages displaying course structure, syllabus, instructor bio, ratings, and student reviews (CourseDetail.jsx).
- Course Enrollment: One-click enrollment creating student progress records (Enrollment.js).
- Interactive Learning Workspace (CourseLearn.jsx):
 - Module and lesson navigation sidebar.
 - YouTube video player integration with real-time video progress tracking.
 - Video scrub/seek prevention enforcing minimum watch percentage before marking lessons complete.
 - Rich text lesson content display with DOMPurify HTML sanitization.
 - Manual lesson completion toggles and live overall progress bar.
- Assignment Workspace (AssignmentView.jsx):

- Detailed assignment briefs, deadlines, and scoring criteria.
- Submission options: Rich text responses, external URLs, or file attachments (uploaded via Multer).
- Status tracking (Pending, Passed, Failed) with instructor scores and feedback.
- Timed Exam & Assessment Space (ExamTake.jsx):
- Countdown timer with automatic submission upon time expiry.
- Supports Multiple Choice (MCQ), Multiple Select, and Free-text/Essay questions.
- Interactive question grid with flagging/bookmarking feature for review.
- Tab Switch Proctoring System: Monitors window blur/tab switching events. Displays warnings and automatically submits the exam after 3 violations.
- Exam Retake Request System: Allows students to request additional exam attempts from instructors when initial attempts are exhausted.
- Certificate Issuance & Verification:
- Automatic PDF certificate generation upon 100% completion of course lessons, assignments, and exams.
- Certificate viewer and PDF download (CertificateView.jsx).
- Public Certificate Verification page (VerifyCertificate.jsx) validating credentials via Certificate ID or QR code URL.
- Student Dashboard (Dashboard.jsx):
- Overview of enrolled courses, completion rates, study time, exam scores, and earned certificates.
- Notifications drawer with read/unread status management.

3. Instructor Features

- Course Builder (CourseCreate.jsx):
- Create and edit course details (Title, Category, Price, Level, Thumbnail, Objectives).
- Module management: Create, update, reorder, or delete modules.
- Lesson management: Create text or video lessons, specify YouTube video IDs, resources, and durations.
- Course status toggle (Draft vs. Published).
- Assignment Builder (AssignmentCreate.jsx):
- Create and edit assignments linked to specific courses.
- Configure passing marks, total marks, due dates, and instructions.
- Assignment Grading Dashboard (GradingDashboard.jsx):
- View all student submissions for a given assignment.
- Inspect student text responses, uploaded files, and link submissions.
- Assign numeric scores, written feedback, and pass/fail statuses.
- Exam Builder (ExamCreate.jsx):
- Create exams with configurable duration, total marks, passing thresholds, and max attempts allowed.
- Interactive Question Builder: Support for single-select MCQs, multi-select MCQs, essay questions, and mark distributions.
- Exam Retake Moderation:
- Review student requests for additional exam attempts.
- Approve or reject retake requests with instructor notes (updateAttemptRequestStatus).
- Instructor Analytics Dashboard:
- Insights on total student enrollments, published courses, active assignments, and pending grading queue.

4. Admin Features

- User Registry & Moderation (AdminPanel.jsx):
- Complete user list displaying user names, emails, roles, and registration dates.
- Filter users by role (student, instructor, admin).
- Delete user accounts with protection preventing accidental self-deletion of active admin accounts.
- System Analytics:
- Real-time platform counts: Total Users, Total Courses, Total Enrollments, and Total Certificates Issued.
- Superuser Privileges:
- Full system-wide access to create, inspect, edit, or delete any course, exam, assignment, or user profile.

5. AI Features (Google Gemini Integration)

- AI Engine (aiService.js / aiController.js):
- Powered by @google/generative-ai configured with the gemini-3.5-flash model.
- Automated Course Description Generator:
- Generates engaging 150-200 word course descriptions based on title and category.
- Automated Lesson Summary Generator:

- Synthesizes lesson text or topics into concise 2-3 sentence summaries.
 - Custom AI Prompt Endpoint:
 - Endpoint (/api/ai/custom) for custom content generation requests.
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6. Technical Features & Backend Architecture

- Database Schemas (Mongoose):
- User, Course, Module, Lesson, Enrollment, Progress, Assignment, Submission, Exam, Question, ExamAttemptRequest, Certificate, Notification, Review.
- Security & Utility Middleware:
- Helmet security headers and CORS cross-origin sharing rules.
- express-rate-limit rate-limiting to defend against DDoS/brute-force attacks.
- Joi payload validation middleware (validate.js) verifying requests prior to controller execution.
- Unified error handling (ApiError.js & asyncHandler.js).
- Resource ownership middleware (checkCourseOwnership.js) ensuring instructors only edit their own assets.
- PDF & QR Code Generation Engine:
- Dynamic PDF certificate generation using PDFKit with custom borders, headers, grades, instructor signatures, and QR codes.
- Server-side QR code generation using qrcode library rendering buffer images pointing to /verify/:certificateId.
- Email Notifications:
- Transactional HTML emails sent via Nodemailer for certificate delivery and notifications.
- File Upload Handler:
- Multer storage engine managing image uploads and assignment attachment files stored in /uploads.